

COLLEGE ERP: DASHBOARD & UI WORKFLOW SPECIFICATION

1. Admin & Principal Web Dashboard

- **Sidebar Navigation:** Sticky left menu featuring 'Data Ingestion', 'Staff Allocation', 'Universal Search', and 'Global Analytics'.
- **Data Ingestion UI:** A central drag-and-drop zone for CSV/Excel uploads to batch-import student lists. Below it, a paginated data table displays the active 7-8 sections across both years.
- **Staff Allocation Matrix:** A grid interface with dropdowns to assign specific Teacher IDs to sections and map them to Session 1, 2, or 3.

2. Teacher Mobile Interface

- **Active Sessions View:** The home screen displays today's assigned sessions as clickable cards (e.g., "Year 1, Sec B - Mid-Day").
- **Rapid Attendance Grid:** Tapping a session opens a list of assigned students. Every student defaults to a green "Present" toggle. Teachers only tap to switch absentees to a red "Absent" state, then press a sticky "Submit" button.
- **Search & Inquiry Tab:** A dedicated bottom-nav tab allowing the teacher to input a Roll Number to instantly access a student's complete historical record.

3. Hostel Warden Mobile Interface

- **Restricted Domain View:** Wardens are hard-locked to their specific assignment (e.g., Boys Hostel, Year 2) upon login.
- **Room-Based Roll Call:** The UI groups the attendance list by physical room numbers or blocks rather than academic sections.
- **Alert Trigger:** Hitting submit automatically dispatches the absence payload to the background SMS gateway to notify parents in real-time.

4. Search & PDF Export Workflow

- **Unified Profile View:** Searching a Roll Number pulls a dashboard merging both class sessions and nightly hostel logs into one chronological timeline.
- **Filter Controls:** Sticky top-bar date pickers allow the teacher or admin to filter this timeline by "Current Month", "Full Year", or a "Custom Date Range".
- **Export & Share:** A prominent "Generate PDF" button compiles the filtered data into a formatted report. Once rendered, it triggers the device's native share sheet.

5. UI Navigation Map (Click-Flow)

This outlines the precise screen-to-screen routing and state changes. The structure is optimized for React Router on the web and Expo stack navigation on mobile.

A. Admin Web Flow

Login Screen → Enter Credentials → Click **Login** → Opens **Admin Dashboard**

Admin Dashboard → Click **Data Ingestion** (Sidebar) → Opens **Upload Portal**

Upload Portal → Click **Choose CSV** → Click **Upload** → Shows **Success Modal** → Redirects to **Sections Table**

Admin Dashboard → Click **Staff Allocation** (Sidebar) → Opens **Allocation Matrix**

B. Teacher Mobile Flow

Login Screen → Enter Credentials → Tap **Login** → Opens **Active Sessions (Home)**

Active Sessions → Tap **Session Card** → Opens **Attendance Grid**

Attendance Grid → Toggle **Present/Absent** → Tap **Submit** → Opens **Confirm Modal** → Tap **Confirm** → Returns to **Active Sessions**

Bottom Tab Bar → Tap **Search/Inquiry** → Opens **Student Search Page**

Student Search Page → Enter Roll No. → Tap **Search** → Opens **Unified Timeline**

Unified Timeline → Tap **Filter** → Select Date → Tap **Generate PDF** → Opens **Share Sheet**

C. Hostel Warden Mobile Flow

Login Screen → Enter Credentials → Tap **Login** → Opens **Hostel Block Dashboard**

Hostel Block Dashboard → Tap **Room Card** → Opens **Room Roll Call List**

Room Roll Call List → Toggle **Present/Absent** → Tap **Submit** → Triggers SMS → Shows **Success Toast** → Returns to **Hostel Block Dashboard**